Hyperbolic Embeddings for Hierarchical Style Representation

Does the geometry of style match the geometry of trees?

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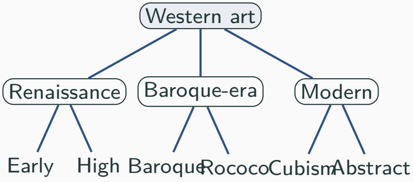
COMPSCI 209B — Spring 2026 / Canvas Project #150

Artistic style is hierarchical

Movements branch along lines of historical influence:

- Renaissance → Baroque → Rococo
- Romanticism → Impressionism → Post-Impressionism
- Cubism → Analytical / Synthetic Cubism

A good style embedding should respect that branching, not just separate classes.



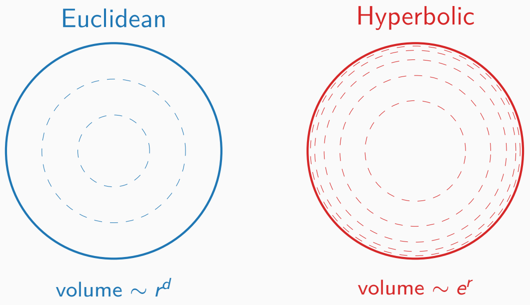
Trees grow exponentially. Euclidean space does not.

Euclidean ball, radius r:
volume grows as r^d *(polynomial)*.

Tree, depth r:
$\#$ nodes grows as b^r *(exponential)*.

$\Rightarrow$ Embedding a deep tree into $\mathbb{R}^d$ must distort distances.

Poincaré ball: hyperbolic, exponential volume growth — a natural home for trees [Nickel & Kiela 2017; Sala et al. 2018].



Equal-distance shells in each geometry. Hyperbolic shells crowd toward the boundary — room for a tree's leaves.

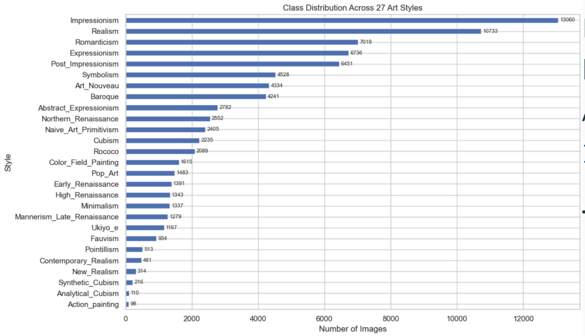
Research question

Do **Hyperbolic** embeddings recover the WikiArt style hierarchy more faithfully than **Euclidean** embeddings of matched dimension and training budget?

We answer along three axes:

1. **Scaling.** Sweep dimension $d \in \{2, 4, 8, 16, 32, 64\}$ and curvature $c \in \{0.1, 0.3, 1, 3\}$.
2. **Training-time hierarchy.** Add a regularizer with weight λ that pulls prototypes toward the tree.
3. **Tree sensitivity.** Re-train against alternative reference trees, including a data-driven empirical one.

Data: WikiArt Refined — 81,446 paintings, 27 styles



Heavy class imbalance

Impressionism: 13,060 images

Action_painting: 98 images

133× ratio between extrema.

Two design choices follow:

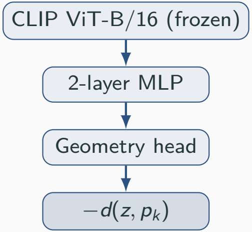
- Report *balanced* accuracy alongside top-1.
- Use a *prototype-softmax* head (no per-class bias).

Splits: supplied 70/30, ∼57k / 24k.

Method: prototype softmax in two geometries

Pipeline

- Frozen CLIP ViT-B/16 features ($\mathbb{R}^{512}$).
- 2-layer MLP $\rightarrow$ embedding $z \in \mathbb{R}^d$.
- Geometry head: identity *or* $\exp_0^c$ onto $\mathbb{B}_c^d$.
- 27 learnable prototypes; logits $= -d(z, p_k)$.



Switching geometry changes only two things: the head's final transform and the distance metric.

Hierarchy-aware loss & training

Loss. Cross-entropy plus a regularizer pulling the prototype distance matrix toward the tree distance matrix:

$$\mathcal{L} = \mathcal{L}_{CE} + \lambda \cdot \frac{1}{|\mathcal{P}|} \sum_{(i,j)\in\mathcal{P}} \left(\frac{d(p_i, p_j)}{\bar{d}} - \frac{T_{ij}}{\bar{T}} \right)^2 .$$

Mean-normalised — constrains the *shape* of $d(p_i, p_j)$, not its scale.

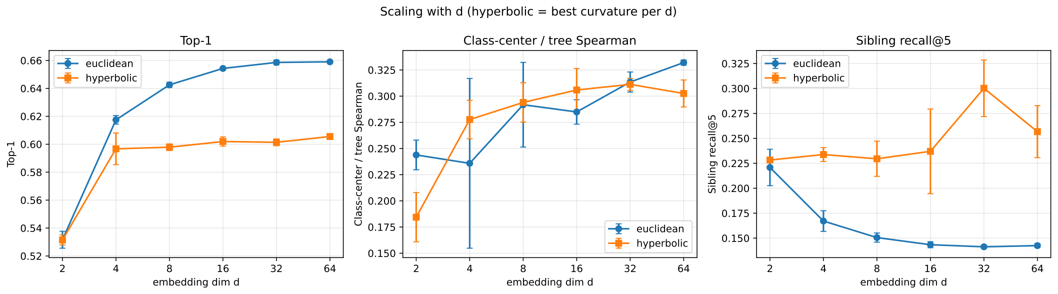
Training

- 150 runs $\times$ 3 seeds; 30 epochs; batch 4096.
- Adam 10^{-3} (head), Riemannian Adam 10^{-2} (**Hyperbolic** prototypes).
- $<$10 s / run; full sweep $<$30 min on one M-series GPU.

Evaluation

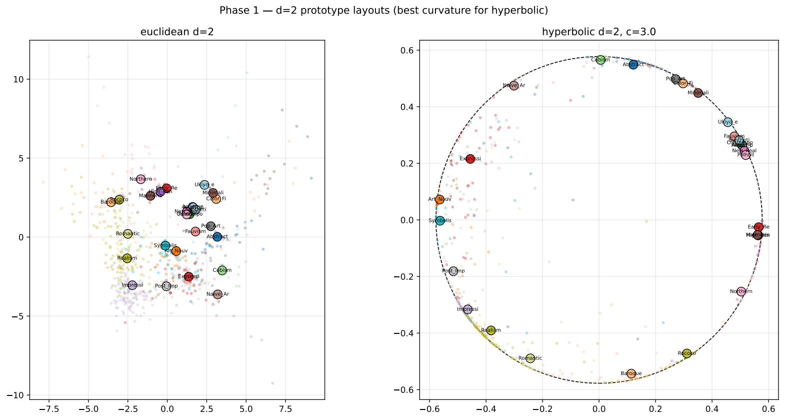
- *Classification:* top-1, top-5, balanced acc.
- *Global tree:* prototype/tree Spearman, distortion.
- *Local hierarchy:* sibling & cousin recall@k.

Phase 1 — scaling sweep: a clean local-vs-global split



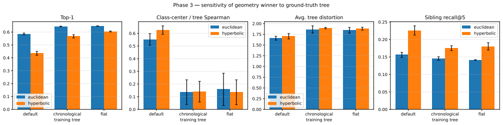
Euclidean leads top-1 by 4–6 pp at every $d \geq 4$. Sibling recall@5 stays flat for **Hyperbolic** but *decreases* with d for **Euclidean** — gap widens to $\sim$5 pp by d=64.

Qualitative: hyperbolic prototypes settle on the boundary



At d=2, **Hyperbolic** prototypes form a near-boundary ring — the exponential-volume regime where leaves should live — while **Euclidean** prototypes scatter without structural pressure.

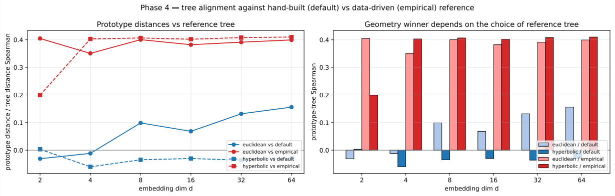
Phases 2–3: regularizer & tree variants don't flip the winner



Phase 2 — sweep λ. Tree-Spearman climbs $0.3 \to 0.7$; classification collapses for $\lambda \geq 1$. Winner unchanged.

Phase 3 — alternative trees. **Euclidean** wins top-1 on every tree; **Hyperbolic** wins sibling recall on every tree.

Phase 4 — against a data-driven tree, the global gap inverts



Empirical tree: agglomerative clustering of CLIP class-mean features.

Hand-built vs. empirical Spearman $= 0.08$ — nearly orthogonal taxonomies.

Against the empirical tree, **Hyperbolic** leads global Spearman at every $d \geq 4$.

The "global **Euclidean** advantage" was an artefact of the hand-built reference.

Takeaways & future work

What we found

- **Hyperbolic** wins *local* hierarchy: sibling recall +4–5 pp. Robust across λ, d, reference tree, class weighting.
- **Euclidean** wins *classification*: top-1 +4–6 pp, equally robust.
- The *global* tree gap depends on the reference — flips in **Hyperbolic**'s favour against a data-driven tree.

Future work

- Fully hyperbolic head (Möbius layers).
- Alternative regularizers: Spearman-rank, Sammon-style.
- Multiple empirical trees to test Phase 4's robustness.

Thank you.

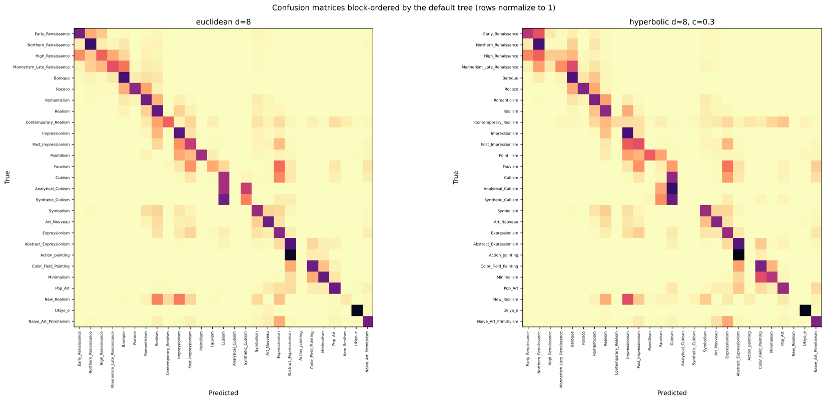
Peter Flo | Luca Grossmann | Valerie Wang

Backup — headline numbers

metric	euclidean	hyperbolic
Top-1	0.659 ± 0.003	0.606 ± 0.002
Top-5	0.959 ± 0.001	0.935 ± 0.001
Balanced acc.	0.554 ± 0.002	0.463 ± 0.004
Class-center / tree Spearman	0.350 ± 0.027	0.242 ± 0.002
Avg. tree distortion	1.742 ± 0.009	1.861 ± 0.003
Worst tree distortion	4.733 ± 0.253	7.147 ± 0.076
Dendrogram F1	0.034 ± 0.030	0.000 ± 0.000
Sibling recall@5	0.136 ± 0.003	0.188 ± 0.004
Cousin recall@5	0.249 ± 0.003	0.296 ± 0.005
Fréchet (Cubism) acc.	0.927 ± 0.029	0.913 ± 0.021

Best **Euclidean** vs. best **Hyperbolic** configuration across Phases 1–2, selected by mean top-1.

Backup — confusion structure (d=8)



Rows/columns ordered by depth-first traversal of the default tree. Both models concentrate mistakes near the diagonal; **Hyperbolic**'s block is softer — more sibling confusion, less long-distance confusion.